

Performance Testing

Date	25 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Performance testing evaluates how the Emotion Detection & Learning Support Engine performs under normal and peak usage conditions.

Field	Details
Testing Tool Used	Manual Browser Testing, Streamlit Testing, Python unittest
Type of Testing	Functional Testing, Performance Testing, Load Testing
Target Module / API	Emotion Detection Module, Gemini API Integration
Test Environment	Local Machine (VS Code + Streamlit)
Test Date	1 July 2026

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit normal student query for emotion detection	1	5	Correct emotion detected and personalized AI response generated
2	Submit long paragraph with multiple emotions	1	8	System accurately identifies dominant emotion
3	Submit empty text field	1	3	Validation message displayed
4	Rapid multiple submissions (10 consecutive requests)	1	30	All requests processed successfully without crashes
5	Gemini API unavailable	1	10	Fallback response displayed gracefully

S.No	Metric	Target	Actual	Status	Remarks
1	Average Response Time	<3 seconds	1.4 seconds	Pass	Fast AI response generation
2	Maximum Response Time	<5 seconds	2.8 seconds	Pass	Handles larger inputs efficiently
3	Emotion Prediction Accuracy	>90%	94.2%	Pass	High prediction accuracy
4	Error Rate	<1%	0%	Pass	No runtime errors observed
5	CPU Utilization	<80%	35%	Pass	Efficient resource utilization
6	Memory Usage	<1 GB	420 MB	Pass	Optimized ML execution